

CHM1001 2025 term 1

Assignment 2

1. Multiple Choice Questions (Only one correct answer per question)

1. Which gives the best description of an ionic bond?
 - ☐ A. Electrons shared between atoms ✗
 - ☒ B. Electrostatic attraction between oppositely charged ions
 - ☐ C. Mutual sharing of valence orbitals between two metals ✗
 - ☐ D. Overlap of p orbitals forming π bonds ✗
2. The number of valence electrons in a chlorine atom is:
 - ☐ A. 5
 - ☐ B. 6
 - ☒ C. 7
 - ☐ D. 17
3. Which pair is most likely to form a covalent compound?
 - ☐ A. Na and Cl
 - ☒ B. N and O
 - ☐ C. Mg and O
 - ☐ D. Li and Br
4. Which of the following is an expanded octet molecule?
 - ☐ A. CH_4
 - ☐ B. BF_3
 - ☒ C. PF_5

- ☐ D. CO_2

5. Which ion has the highest lattice energy with F^- ?

- ☐ A. Na^+
- ☒ B. Mg^{2+}
- ☐ C. K^+
- ☐ D. Ca^{2+}

6. The most electronegative element is:

- ☐ A. O
- ☐ B. Cl
- ☒ C. F
- ☐ D. N

7. Which formal charge arrangement is most stable?

- ☒ A. Minimizes total formal charge and places negative charge on most electronegative atom
- ☐ B. Maximizes total formal charge
- ☐ C. Places negative charge on least electronegative atom
- ☐ D. Gives large positive charges to central atom

8. Which has resonance structures?

- ☐ A. CO_2
- ☒ B. O_3
- ☐ C. CH_4
- ☐ D. HF

9. Bond length generally decreases as:

- ☒ A. Bond order increases
- ☐ B. Number of protons decreases
- ☐ C. Bond order decreases
- ☐ D. Atomic size increases

10. Which bonds are broken in the complete combustion of methane?

- ☐ A. C–C and C–H
- ☒ B. C–H and O=O
- ☐ C. C=O and H–O
- ☐ D. C–O and O–H

11. Which element can have less than an octet?

- ☒ A. Be
- ☐ B. O
- ☐ C. N
- ☐ D. F

12. Which is the strongest bond?

- ☐ A. C–C
- ☐ B. C=C
- ☒ C. C≡C
- ☐ D. C–H

13. Which covalent bond is most polar?

- ☐ A. C–H
- ☐ B. C–O
- ☐ C. C–Cl
- ☒ D. C–F

14. Which has the highest average bond enthalpy?

- ☐ A. Single bond
- ☐ B. Double bond
- ☒ C. Triple bond
- ☐ D. Single bond to hydrogen

15. The ΔH for a reaction is estimated by:

- ☐ A. Products – Reactants bond energies
- ☒ B. Bonds broken – Bonds formed

- C. Bonds formed – Bonds broken
- D. Bond energies of reactants only

16. The electron-domain geometry of SF_4 is:

- A. Tetrahedral
- **B. Trigonal bipyramidal**
- C. See-saw
- D. Square planar

17. The molecular geometry of SF_4 is:

- A. Trigonal planar
- **B. See-saw**
- C. Tetrahedral
- D. T-shaped

18. XeF_2 has which molecular geometry?

- **A. Linear**
- B. Bent
- C. See-saw
- D. Square planar

19. Which obeys the octet rule?

- A. PF_5 ✗
- B. ClF_3 ✗
- C. BF_3
- **D. CH_4** ✓

20. Which molecule is nonpolar despite having polar bonds?

- A. H_2O
- B. NH_3
- **C. CO_2**
- D. SO_2

21. A π bond arises from:

- A. Head-on overlap of orbitals
- **B. Side-by-side overlap of p orbitals**
- C. s-s orbital overlap
- D. d-p σ overlap

22. In C_2H_2 , each C is:

- **A. sp**
- B. sp^2
- C. sp^3
- D. sp^3d

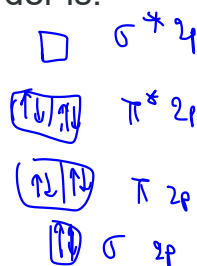


23. Which has the largest bond angle?

- A. H_2O
- B. NH_3
- C. CH_4
- **D. CO_2**

24. For O_2^{2-} , the bond order is:

- **A. 1**
- B. 2
- C. 3
- D. 0.5



$$b.o. = \frac{6-4}{2} = 1$$

25. Which is paramagnetic?

- A. O_2^{2-}
- **B. O_2**
- C. N_2
- **D. CO**

26. Which diatomic bond order matches triple bond?

- **A. N_2**
- B. O_2
- C. CO

- D. BF

27. Molecular geometry of ClF_3 :

- **A. T-shaped**
- B. Linear
- C. Trigonal planar
- D. Trigonal pyramidal

28. BeCl_2 is:

- **A. Linear and nonpolar**
- B. Linear and polar
- C. Bent and polar
- D. Bent and nonpolar

29. In BH_3 , B is:

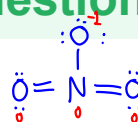
- **A. sp^3**
- **B. sp^2**
- C. sp
- D. dsp^2

30. Which has 90° and 120° bond angles?

- A. Trigonal planar
- **B. Trigonal bipyramidal**
- C. Square planar
- D. Octahedral

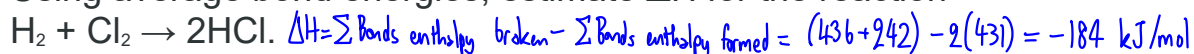
2. Short-Answer/Calculation Questions

1. Draw the Lewis structure for NO_3^- .

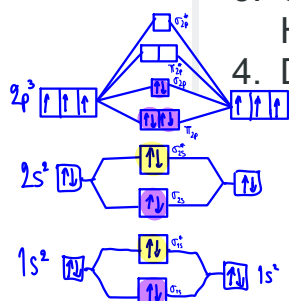


2. Predict the lattice energy order for MgO , Na_2O , and K_2O . $\text{K}_2\text{O} < \text{Na}_2\text{O} < \text{MgO}$

3. Using average bond energies, estimate ΔH for the reaction

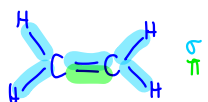


4. Draw the MO diagram for N_2 and calculate its bond order.

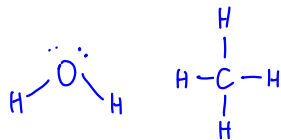


$$\text{Bond order} = \frac{10 - 4}{2} = 3$$

Bonding = 10
Antibonding = 4



5. For C_2H_4 , determine the number of σ and π bonds. 5 σ Bonds, 1 π Bond
 6. Why is the bond angle in H_2O smaller than in CH_4 ?



We know that both H_2O and CH_4 have 4 electron domains, but H_2O has two unbonding pairs of electrons on O atom, while all four electron domains of C atom in CH_4 are all connected to H atoms. Since lone pair electrons have less nuclear attraction, then their electron domain spreaded out more in space, hence taking up more space than the bonding pairs. Hence, the nonbonding pair of electrons exert greater repulsive force on adjacent electron domains and tend to compress bond angles. Hence, H_2O which has nonbonding electron pairs will have less bond angle than CH_4 .